

AGILE SPRINT PLANNING & USER STORY WORKSHOP

CO1 – Assessment Task 2

Problem Statement #37

AI-Based Digital Waste Recycling Marketplace and Resource Recovery System

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Framework Applied: Scrum / Agile Software Engineering Practices

Table of Contents

1. Introduction	3
2. Project Vision & Stakeholder Identification	3
2.1 Project Objectives	3
2.2 Key Stakeholders	4
2.3 System Overview	5
3. Epics.....	5
4. Epics & User Stories.....	6
5. Acceptance Criteria	7
6. Product Backlog Prioritization	9
7. Sprint 1 Backlog & Task Breakdown	10
7.1 Sprint Framework	10
7.2 Sprint 1 Selected Stories	11
8. Story Point Estimation (Planning Poker).....	12
9. Sprint Goal & Expected Deliverables	13
9.1 Sprint 1 Goal	13
9.2 Sprint 1 Deliverables	13
9.3 How Sprints Roll Up to the Project's Expected Outcomes.....	14
9.4 Definition of Done	14

1. Introduction

Rapid urbanization has sharply increased both municipal and industrial waste generation, while existing waste-management systems remain fragmented and poorly coordinated among the parties who generate, collect, recycle, and trade recyclable materials. Problem Statement #37 calls for an AI-powered digital marketplace that connects households, industries, recyclers, and collection agencies on a single platform — one that automatically classifies waste, recommends recycling centres, optimises collection schedules, facilitates trading of recyclable materials, and generates sustainability analytics to strengthen circular-economy outcomes.

This document applies Agile software engineering practices to translate that problem statement into an actionable, sprint-ready plan. It defines the project vision and stakeholders, decomposes the requirements into Epics and User Stories, specifies measurable acceptance criteria, prioritises the product backlog, plans the first sprint in detail with task-level breakdown and Fibonacci-based story-point estimation, and defines the Sprint Goal together with the expected deliverables.

2. Project Vision & Stakeholder Identification

Project Vision Statement

For households, industries, recyclers, and collection agencies who currently coordinate recyclable-waste trading through disconnected, manual channels, the AI-Based Digital Waste Recycling Marketplace is a cloud- and IoT-enabled platform that automatically classifies waste, matches generators with the right recyclers, optimises collection routes, and generates sustainability analytics. Unlike traditional waste-management systems that operate in silos with poor visibility, our platform unifies discovery, trading, logistics, and reporting into one transparent marketplace — maximising resource recovery, reducing landfill dependency, and accelerating circular-economy adoption.

2.1 Project Objectives

- Digitize recyclable waste trading.
- Classify waste materials using AI.
- Connect waste generators with recyclers.
- Optimize waste collection schedules.
- Monitor recycling activities.
- Generate sustainability analytics.
- Promote circular economy practices.

- Reduce landfill waste.

2.2 Key Stakeholders

Stakeholder	Role / Description	Primary Interest
Household Waste Generators	Everyday citizens who list recyclable household waste for pickup or sale.	Ease of listing; fair pricing; convenient scheduling.
Industrial / Commercial Waste Generators	Factories and businesses producing bulk recyclable waste streams.	Bulk trading, compliance, verified buyers.
Waste Collection Agencies	Municipal or private agencies that pick up listed waste.	Optimized routes, reduced fuel/time cost.
Recycling Facilities / Recyclers	Process recyclable material sourced through the marketplace.	Reliable, well-classified material supply.
Buyers of Recyclable Materials	Industries purchasing processed recycled raw material.	Trust, quality assurance, transparent trading.
Platform Administrators / Operations Team	Manage platform operations, disputes, and quality control.	Full visibility and control over the marketplace.
Environmental Regulatory Bodies	Government / Pollution Control Boards overseeing compliance.	Auditable recycling records and compliance data.
Sustainability Officers (Municipal / CSR)	Track circular-economy KPIs for their organisation or city.	Reliable dashboards and impact reporting.
AI/ML & Platform Development Team	Build and maintain classification models and platform features.	Clear requirements, clean data pipelines.
Logistics / Delivery Partners	Execute physical pickup and transport of waste.	Clear schedules and route information.

2.3 System Overview

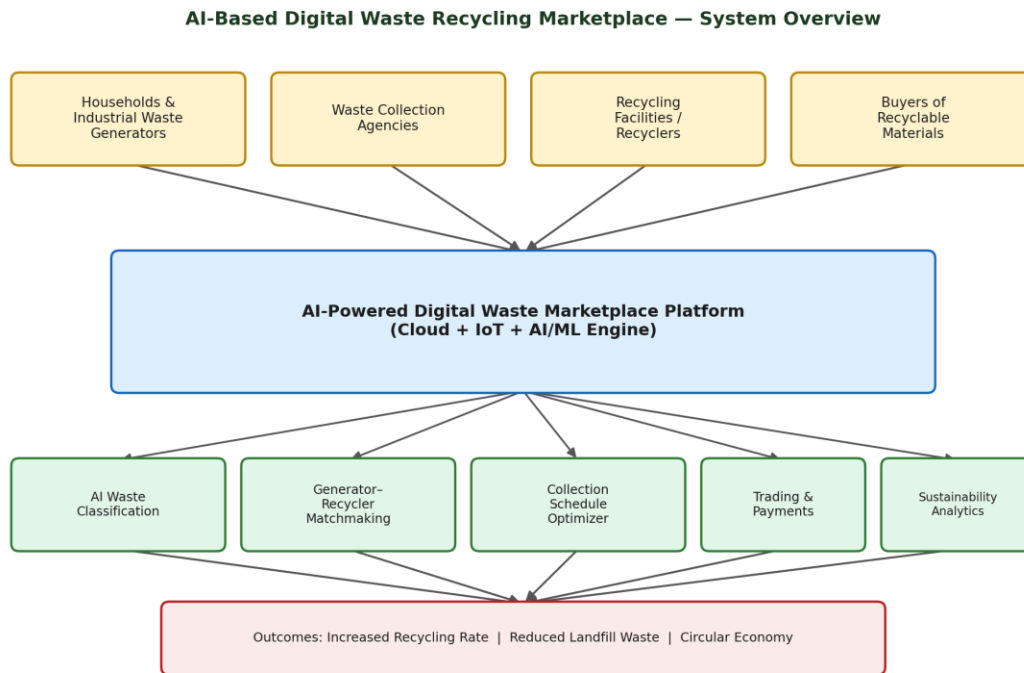


Figure 1: High-level architecture showing stakeholders, core platform modules, and target outcomes.

3. Epics

The eight project objectives were grouped into eight Epics — large bodies of related functionality that are each broken down into smaller User Stories in Section 4.

Epic ID	Epic Name	Linked Objective(s)
E1	Digitized Waste Trading & Marketplace	Objectives 1, 7
E2	AI-Based Waste Classification	Objective 2
E3	Generator–Recycler Connection & Matchmaking	Objective 3
E4	Waste Collection Schedule Optimization	Objective 4
E5	Recycling Activity Monitoring & Compliance	Objectives 5, 8
E6	Sustainability Analytics & Reporting	Objectives 6, 7
E7	User Onboarding & Account Management	Enabling epic (all objectives)
E8	Community Engagement & Circular-Economy Incentives	Objectives 7, 8

4. Epics & User Stories

Each Epic was broken down into well-structured User Stories following the standard Agile format: “As a <user>, I want <feature>, so that <benefit>.” Each story is tagged with a MoSCoW priority used later for backlog ordering.

E1 — Digitized Waste Trading & Marketplace

ID	User Story	Priority
US1	As a household waste generator, I want to list my recyclable waste items with photos and quantity, so that recyclers can find and buy them easily.	Must
US2	As an industrial waste generator, I want to post bulk waste lots with material specifications, so that I can trade at better prices with verified recyclers.	Must
US3	As a recycler, I want to browse and filter available waste listings by material type, quantity, and location, so that I can source raw material efficiently.	Must
US4	As a buyer, I want to place an order and make a secure payment for recyclable material, so that I can complete purchase transactions on the platform.	Must

E2 — AI-Based Waste Classification

ID	User Story	Priority
US5	As a waste generator, I want the system to automatically classify my uploaded waste photo into a material category, so that I do not need manual sorting expertise.	Must
US6	As a recycling facility, I want AI to flag contaminated or mixed waste before pickup, so that I can request re-sorting or reject the listing early.	Should

E3 — Generator–Recycler Connection & Matchmaking

ID	User Story	Priority
US7	As a waste generator, I want to see nearby recycling centres recommended based on my waste type, so that I can choose the most convenient option.	Must
US8	As a recycler, I want to receive automatic notifications of matching waste listings within my service radius, so that I can respond quickly to new supply.	Should

E4 — Waste Collection Schedule Optimization

ID	User Story	Priority
US9	As a waste collection agency, I want an optimised pickup route and schedule generated for multiple listings, so that I can reduce travel time and fuel cost.	Should

ID	User Story	Priority
US10	As a household user, I want to book or reschedule a pickup slot, so that I can choose a time that is convenient for me.	Must

E5 — Recycling Activity Monitoring & Compliance

ID	User Story	Priority
US11	As a platform administrator, I want to track the status of each waste item from listing to recycling completion, so that I can ensure accountability across the chain.	Should
US12	As a regulatory body, I want access to verified records of recycled quantities per recycler, so that I can audit compliance with recycling norms.	Could

E6 — Sustainability Analytics & Reporting

ID	User Story	Priority
US13	As a municipal sustainability officer, I want dashboards showing recycling rates, emissions saved, and landfill diversion, so that I can report progress on circular-economy goals.	Should
US14	As a recycler, I want a monthly sustainability report of materials processed, so that I can showcase my environmental impact to clients.	Could

E7 — User Onboarding & Account Management

ID	User Story	Priority
US15	As a new user, I want to register and verify my profile as a household, industry, recycler, or agency, so that I can access role-specific features.	Must

E8 — Community Engagement & Circular-Economy Incentives

ID	User Story	Priority
US16	As a household user, I want to earn eco-points or rewards for consistent recycling, so that I stay motivated to participate.	Could

5. Acceptance Criteria

Acceptance criteria are written in Given–When–Then format so that each story has a clear, testable definition of “done”. The five Sprint 1 stories (US15, US1, US3, US5, US7) each include an additional edge-case criterion for extra rigor.

Story	Given	When	Then
US1	a household user is creating a new waste listing	they upload at least one photo and enter waste type and quantity	the listing is saved, marked “Pending Classification”, and visible to matching recyclers.
US1	a listing is missing a required field (photo, type, or quantity)	the user attempts to submit it	the system blocks submission and highlights the missing field.
US2	an industrial user is posting a bulk waste lot	they specify material grade, quantity, and preferred trading price	the lot is published to verified industrial buyers with a “Bulk” tag.
US3	a recycler opens the marketplace search	they filter by material type, location radius, and quantity	only matching, currently available listings are returned, sorted by distance.
US4	a buyer selects a listing and proceeds to checkout	they confirm quantity and complete payment	an order confirmation is generated and the listing status changes to “Sold”.
US5	a waste photo is uploaded	the AI classification service processes the image	a material category (e.g., plastic, e-waste, metal, organic) is returned within 5 seconds with a confidence score.
US5	the AI confidence score is below 60%	classification completes	the listing is flagged “Needs Manual Review” instead of being auto-published.
US6	a listing photo shows mixed or contaminated material	the AI model evaluates it	the listing is flagged “Contamination Risk” and the recycler is notified before pickup.
US7	a listing is successfully classified	the generator views the listing detail page	the three nearest suitable recycling centres are shown, ranked by distance and acceptance of that material type.
US8	a new listing matches a recycler's registered material preferences and radius	the listing is published	the recycler receives an in-app and email notification within 2 minutes.
US9	multiple pickups are scheduled for a collection agency on the same day	the agency requests a route plan	the system returns an optimised sequence of stops minimising total travel distance.
US10	a household user has a pending pickup	they select a new date/time from available slots	the schedule updates and both parties receive a confirmation notification.

Story	Given	When	Then
US11	a waste item has moved between statuses (Listed → Matched → Collected → Recycled)	an administrator opens the tracking dashboard	the current status and full status history are displayed for that item.
US12	a regulator requests recycling records for a specific recycler and date range	they query the compliance module	a verified, exportable report of recycled quantities is generated.
US13	a sustainability officer opens the analytics dashboard	they select a date range and region	recycling rate, estimated emissions avoided, and landfill diversion percentage are displayed as charts.
US14	a calendar month ends	the reporting job runs	each recycler automatically receives a PDF summary of materials processed that month.
US15	a new user signs up	they select a role (household / industry / recycler / agency) and verify via OTP	a role-specific account is created and the user is directed to the relevant dashboard.
US16	a household user completes a recycling transaction	the transaction is marked complete	eco-points are credited to their account and reflected in their profile.

6. Product Backlog Prioritization

The backlog was prioritised using a combination of business value, dependency order, and MoSCoW ranking. Foundational capabilities that unlock everything else — account onboarding, waste listing, AI classification, and discovery — were ranked highest because later features (trading, scheduling, analytics) all depend on a working listing-and-classification core. Monetisation features (bulk trading, payments, scheduling) form the next tier, followed by monitoring, and finally analytics and engagement features that add strategic value once the core marketplace is operating.

Rank	ID	User Story (Summary)	Epic	MoSCoW	Value	Pts	Sprint
1	US15	Register & verify role-based account	E7	Must	High	3	Sprint 1
2	US1	List recyclable waste item (household)	E1	Must	High	5	Sprint 1
3	US3	Browse & filter waste listings (recycler)	E1	Must	High	5	Sprint 1
4	US5	AI auto-classification of waste photo	E2	Must	High	13	Sprint 1

Rank	ID	User Story (Summary)	Epic	MoSCoW	Value	Pts	Sprint
5	US7	Nearby recycling-centre recommendation	E3	Must	High	8	Sprint 1
6	US2	Post bulk industrial waste lot	E1	Must	High	8	Sprint 2
7	US4	Order & secure payment	E1	Must	High	8	Sprint 2
8	US10	Book / reschedule pickup slot	E4	Must	High	5	Sprint 2
9	US6	AI contamination flagging	E2	Should	Medium	8	Sprint 2
10	US8	Notify recycler of matching listing	E3	Should	Medium	5	Sprint 2
11	US9	Optimized pickup route & schedule	E4	Should	Medium	13	Sprint 2
12	US11	Track item status end-to-end	E5	Should	Medium	8	Sprint 2
13	US13	Sustainability analytics dashboard	E6	Should	Medium	8	Sprint 3
14	US12	Regulatory audit records	E5	Could	Low	5	Sprint 3
15	US14	Monthly sustainability report (recycler)	E6	Could	Low	5	Sprint 3
16	US16	Eco-points / rewards	E8	Could	Low	5	Sprint 3

Prioritization rationale: Stories 1–5 form a minimum viable product (list → classify → discover) and are scheduled for Sprint 1. Stories 6–12 extend the marketplace into full trading and logistics for Sprint 2. Stories 13–16 add compliance, analytics, and engagement value in Sprint 3, once real transaction data exists to analyse.

7. Sprint 1 Backlog & Task Breakdown

7.1 Sprint Framework

Sprint duration: 2 weeks (10 working days). Team: a 6-member cross-functional squad (1 Scrum Master / Product Owner, 2 Backend Developers, 1 AI/ML Engineer, 1 Frontend Developer, 1 QA Engineer).

Assumed team velocity: ~35 story points per sprint, based on total sprint capacity (6 members × ~6 focused hours/day × 10 days ≈ 360 hours) and a benchmark of roughly 10 effort-hours per story point once ceremonies, code review, and integration buffer are accounted for.

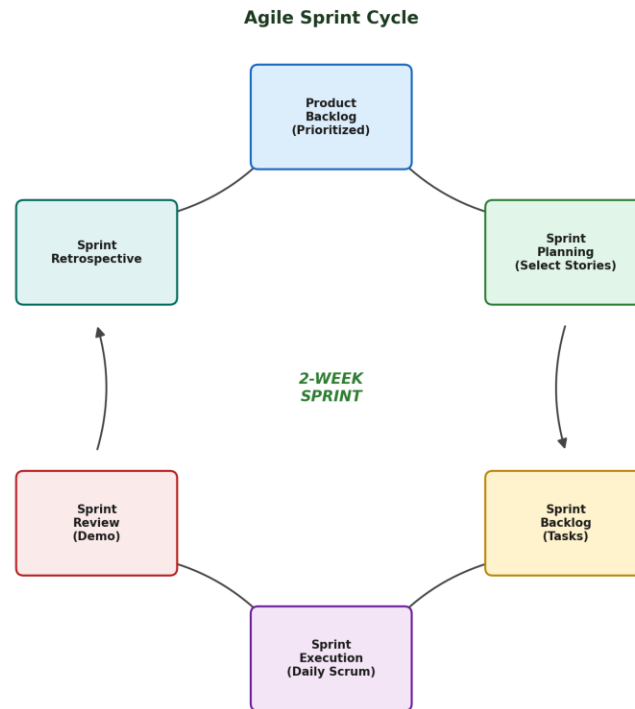


Figure 2: The recurring Scrum sprint cycle applied to this project.

7.2 Sprint 1 Selected Stories

The five highest-priority stories (34 story points, within the ~35-point velocity budget) were selected for Sprint 1 because together they deliver one complete, demonstrable user journey: a user registers, lists waste, the platform classifies it with AI, and a recycler discovers it through search or recommendation.

Story	Task	Owner	Est. Hours
US15 Register & verify account (3 pts)	Design signup/login UI with role selector	Frontend Dev	6h
	Build role-based registration API	Backend Dev	8h
	Implement OTP/email verification	Backend Dev	6h
	Write unit & UI tests	QA	4h
US1 List recyclable waste item (5 pts)	Design “Add Listing” form with photo upload	Frontend Dev	8h
	Build listing creation API & DB schema	Backend Dev	10h
	Integrate cloud image storage	Backend Dev	6h

Story	Task	Owner	Est. Hours
	Test listing CRUD & photo upload	QA	6h
US3 Browse & filter listings (5 pts)	Build search/filter UI (type, location, qty)	Frontend Dev	8h
	Implement filter/search API with indexing	Backend Dev	10h
	Optimize geo-filter query performance	Backend Dev	6h
	Test filter combinations	QA	6h
US5 AI auto-classification (13 pts)	Curate & label waste-image training dataset	AI/ML Engineer	16h
	Train & validate classification model	AI/ML Engineer	20h
	Build classification microservice/API	Backend Dev	10h
	Integrate classification into listing flow	Full-stack Dev	8h
	Accuracy testing on sample image set	QA	8h
US7 Nearby recycling-centre recommendation (8 pts)	Implement geolocation-based recommendation logic	Backend Dev	10h
	Build “Recommended Centres” UI component	Frontend Dev	8h
	Integrate maps API for distance calculation	Backend Dev	8h
	Test recommendation accuracy across locations	QA	6h

Total planned effort: ~178 hours against a ~360-hour team capacity, leaving buffer for daily stand-ups, sprint ceremonies, code review, and unplanned bug fixes — keeping the sprint backlog realistic rather than over-committed.

8. Story Point Estimation (Planning Poker)

Story points were estimated using Planning Poker with a Fibonacci-like scale (1, 2, 3, 5, 8, 13, 21), which widens the gaps between larger numbers to reflect growing uncertainty in bigger stories. Each team member privately selected a card representing their estimate; estimates were revealed simultaneously, and any large disagreement (e.g., a 3 versus a 13) triggered a short discussion where the highest and lowest estimators explained their reasoning before the team re-voted and converged on a shared value.

Story	Points	Justification
US15	3	Simple CRUD-style registration flow with an established OTP pattern; low technical uncertainty.
US1	5	Standard form + file-upload feature; moderate complexity from image storage integration.
US2	8	Similar to US1 but adds bulk-lot specifications and pricing logic for industrial users.
US3	5	Search/filter logic is well-understood; some effort needed for geo-based query performance.
US4	8	Payment integration introduces external-gateway dependency and security/testing overhead.
US5	13	Highest complexity: requires dataset curation, model training/validation, and accuracy tuning — high technical uncertainty warrants the largest Fibonacci step.
US6	8	Builds on the US5 model but adds a new contamination-detection use case and notification logic.
US7	8	Requires geolocation logic and maps-API integration on top of classification output.
US8	5	Notification logic on an existing matching engine; moderate, well-scoped effort.
US9	13	Route optimisation across multiple stops is algorithmically complex and needs real-world testing.
US10	5	Standard scheduling/calendar feature with moderate UI and notification work.
US11	8	Requires a status-tracking data model spanning multiple services end-to-end.
US12	5	Read-only reporting on existing tracked data; low-to-moderate effort.
US13	8	Dashboard requires aggregation logic across multiple data sources and chart rendering.
US14	5	Scheduled report generation reusing the US13 aggregation logic; smaller incremental effort.
US16	5	Simple points-ledger feature; low complexity but needs profile and transaction hooks.

9. Sprint Goal & Expected Deliverables

9.1 Sprint 1 Goal

“By the end of Sprint 1, enable a complete core user journey in which a registered waste generator can list recyclable waste — automatically classified by AI — and a recycler can discover that listing through search, filters, and location-based recommendations — establishing the foundational trading and intelligence layer of the marketplace.”

9.2 Sprint 1 Deliverables

- A working, role-based registration and login module (household / industry / recycler / agency).
- A waste-listing feature supporting photo upload, quantity, and material description.
- An AI-based waste-classification microservice returning a material category and confidence score.
- A search-and-filter module allowing recyclers to discover listings by type, location, and quantity.
- A location-based “recommended recycling centres” feature.
- A demo-ready increment presented at the Sprint Review, with automated tests covering all five stories.

9.3 How Sprints Roll Up to the Project's Expected Outcomes

Expected Outcome (from Problem Statement)	Delivered By	How
Increased recycling rates	Sprint 1	AI classification + discovery makes it effortless to list and find recyclable material.
Reduced environmental pollution	Sprint 2–3	Enabled once trading volume and monitoring data begin flowing.
Improved waste collection efficiency	Sprint 2	Delivered via optimised pickup scheduling (US9, US10).
Better resource recovery	Sprint 1–2	Direct result of accurate classification and matchmaking.
Enhanced stakeholder collaboration	Sprint 1–2	Marketplace connects generators, recyclers, agencies, and buyers directly.
Reduced landfill dependency	Sprint 2–3	Tracked and reported via monitoring and analytics features.
Sustainable waste management	Sprint 3	Delivered through sustainability dashboards and compliance reporting.
Increased public participation in recycling	Sprint 1, 3	Easy listing (Sprint 1) plus eco-point incentives (Sprint 3) drive adoption.

9.4 Definition of Done

- Code peer-reviewed and merged to the main branch.
- Unit and integration tests written and passing (≥ 80% coverage on new code).
- Feature verified against its acceptance criteria by QA.
- UI reviewed for basic accessibility and responsiveness.
- Feature demonstrated successfully in the Sprint Review.